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Central/Eastern Indian Bundelkhand and Bastar cratons in the Palaeoproterozoic supercontinental reconstructions: A palaeomagnetic perspective

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ABSTRACT

Mafic dykes in the central Indian Bundelkhand and eastern Indian Bastar cratons are potential sources for tracing the location of the Indian shield within Palaeoproterozoic supercontinent reconstructions. A total of 610 oriented core samples were collected from 27 mafic dykes (342 samples) in Bundelkhand craton and 20 dykes (268 samples) in Bastar craton. The characteristic magnetisations identify distinct groups of directions. The derived palaeopoles are integrated with recently reported U–Pb isotopic ages from the Bundelkhand and Bastar cratons and correlated with palaeopoles from the mafic dykes in the Dharwar craton, South India. Characteristic remanence (ChRM) identified in these dykes are classified into (i) steep upward/downward components further sub-grouped as ca. 2.37 Ga steep 1 ($\lambda = 10.2^\circ$S; $\Phi = 75.0^\circ$E; $A_{95} = 18.1^\circ$ in Bundelkhand and $\lambda = 22.3^\circ$N; $\Phi = 71.4^\circ$E; $A_{95} = 21.6^\circ$ in Bastar), component comparable with ca. 2.4–2.45 Ga steep 2 group from one dyke in Bundelkhand ($\lambda = 14^\circ$S; $\Phi = 101^\circ$E; $A_{95} = 26.1^\circ$) and one dyke in Bastar ($\lambda = 6^\circ$N; $\Phi = 113^\circ$E; $A_{95} = 26.5^\circ$). A steep component ($\lambda = 60.4^\circ$N; $\Phi = 45.3^\circ$E; $A_{95} = 9.7^\circ$ in Bundelkhand and a comparable component $\lambda = 49^\circ$N; $\Phi = 129^\circ$E; $A_{95} = 15.1^\circ$ from one dyke in Bastar) is not assigned an age at present. (ii) ca. 2.18 Ga shallow easterly and antipodal shallow westerly components ($\lambda = 0.4^\circ$S; $\Phi = 347^\circ$E; $A_{95} = 21.6^\circ$ in Bundelkhand and $\lambda = 18.0^\circ$N; $\Phi = 344.0^\circ$E; $A_{95} = 8.1^\circ$ in Bastar) and (iii) 1.99 Ga shallow northwest and antipodal shallow southeast ($\lambda = 57.5^\circ$N; $\Phi = 309.0^\circ$E; $A_{95} = 4.7^\circ$ in Bundelkhand and $\lambda = 39^\circ$N; $\Phi = 321^\circ$E; $A_{95} = 28^\circ$ in Bastar). A group (iv) of ~2.2 Ga northeast shallow components ($\lambda = 36.0^\circ$S; $\Phi = 357.0^\circ$E; $A_{95} = 9.4^\circ$) is found only in the Bundelkhand craton. The distinct groups of palaeomagnetic pole determinations from dykes of the Bundelkhand and Bastar craton exhibit a remarkable match with palaeomagnetic poles determined from Precambrian mafic dykes in the Dharwar craton. The close comparison of mafic dyke magnetisations between the cratons suggests close proximity since 2.45–2.5 Ga. Models suggesting amalgamation of crustal blocks along the Central Indian Tectonic Zone at 1.8 Ga or a 1.0 Ga collision along this zone to form Rodinia are untenable. Testing of proposed NeoArchaean-Palaeoproterozoic supercontinent reconstructions showing a north China–India linkage or India's close proximity to Slave craton to form a supercraton 'Sclavia' are not supported. Instead, the data are compatible placing India in close proximity to the Yilgarn block of Western Australia.

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1. Introduction

Following Wegener's hypothesis of Pangea and advancement of the plate tectonic hypothesis for evolution of late Phanerozoic distribution of continental fragments, Piper (1982) and Anderson (1982) have visualized the existence of supercontinent as long ago as the Palaeoproterozoic. However, only in 1990s there has been a growing interest on the pre-Pangea continental configurations, but it has been mostly focused on the Neoproterozoic continental assembly described in literature as "Rodinia" (McMenamin and MacMenamin, 1990) and the Rodinia-Gondwana transition. This has generated considerable interest in palaeomagnetic studies of Precambrian rocks particularly because palaeomagnetic studies uniquely provide quantitative information on the spatial proximity of the continental blocks in contrast to geological correlations which are non-unique. Much of the palaeomagnetic record along with high precision geochronological studies has been generated for Neoproterozoic times to test the Rodinia configuration and the Meso- and Palaeoproterozoic configurations remain largely untested quantitatively. Nonetheless, studies by many Precambrian geologists have brought into light the existence of 2.3–1.8 Ga old orogenic belts and accretionary crustal growth across widely dispersed continental blocks (Rogers, 1996; Condie, 1998; Kusky et al., 2007; French and Heaman, 2010; Söderlund et al., 2010 and

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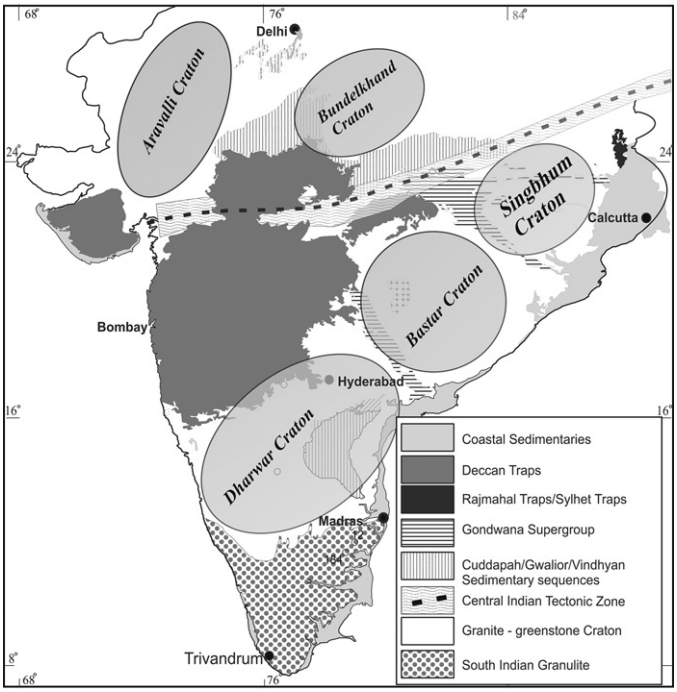


Fig. 1. Broad tectonic sketch of the Indian shield showing the distribution of Archaean cratons and other major geological features.

references therein). These studies contribute towards proposals for a Meso- and Palaeoproterozoic supercontinent configuration, although the history of this proposed supercontinent remains unclear. Recent reviews giving an account of these aspects include Bleeker (2003) and Zhao et al. (2004).

Although a Neoarchaean supercontinent named Kenorland was proposed (Williams et al., 1991), as there are no compelling evidences to support the notion, Bleeker (2003) visualized that many of the Archaean cratons represent fragments of a larger crust–mantle system – a supercraton. The supercratons are postulated to have existed across the Archaean-Proterozoic boundary from the time of amalgamation at ca. 2.6 Ga. Early Proterozoic (2.45–2.10 Ga) is marked by protracted history of diachronous break-up of these cratons, drifting of fragments independently and coalescence along the globally distributed 2.1–1.8 Ga collisional orogens forming a supercontinent that is variably described as Hudsonland, Nuna and Columbia (Condie, 2002; Meert, 2002; Rogers and Santosh, 2002; Rao and Reddy, 2002; Zhao et al., 2004; Wingate et al., 2009; Evans and Mitchell, 2011; Zhang et al., 2012 and references therein). One of the Neoarchaean supercratons proposed is "Sclavia" in which Dharwar craton in India, Zimbabwe and Wyoming are suggested as the nearest neighbours to the Slave craton (Bleeker, 2003). Dyke swarms are considered to provide robust piercing points in palaeocontinental reconstructions although they may be subsequently fragmented and partially destroyed and dispersed globally. French et al. (2008) and French and Heaman (2010) used U–Pb baddeleyite and zircon dating of mafic dykes in the Indian shield to test the juxtaposition of Dharwar and Slave cratons. Similarly, in the Columbia configuration, the Indian shield is shown adjacent to north-China based on correlation of the supposed 1.8 Ga orogens (Zhao et al., 2003). Palaeomagnetic investigations coupled with U–Pb baddeleyite/zircon dating from the Precambrian cratons have recently attained significance to provide better insight into these models.

The Indian shield comprises some of the most ancient cratonic blocks (Fig. 1) and includes the Dharwar craton in the south, the Bastar and Singbhum cratons in the east and the Bundelkhand and Aravalli cratons in the north. Collectively these make the Indian shield an obvious choice for tracing early NeoArchean-Palaeoproterozoic supercontinental history. The Dharwar craton has been central to many geological and palaeomagnetic studies, but similar interest has been lacking for the Bundelkhand and Bastar cratons. Mafic dykes of Proterozoic age occur profoundly in all Precambrian cratons and are potential targets for palaeomagnetic studies. We have carried out such studies on a large number of mafic dykes in India (Radhakrishna and Joseph, 1996a,b; Radhakrishna et al., 2003). Other major studies include studies by Hargraves and Bhalla (1983 and references therein), Venkatesh et al. (1987), Dawson and Hargraves (1994) and Halls et al. (2007). The results of our study on several dykes around the Cuddapah basin in the Dharwar craton are the subject of an accompanying paper (Radhakrishna et al., submitted for publication). This present paper presents results from mafic dykes in Bundelkhand and Bastar cratons. We also note a simultaneous study on the mafic dykes in Bundelkhand (Pradhan et al., 2012) and Bastar (Meert et al., 2011) cratons and combine these results with our study to discuss implications for the geotectonic framework of the Indian shield and supercontinent history during the Palaeoproterozoic.

2. Geological setting

Both the Bundelkhand and Bastar cratons comprise granitic gneiss complexes of Archaean age. The basement of Bundelkhand craton comprises ~3.5 Ga highly deformed gneiss-greenstone assemblages and 2.5 Ga granite plutons (Basu, 1986; Sharma and Rahman, 2000). Numerous quartz veins represent hydrothermal activity that is profusely distributed mainly in NNE–SSW and NE–SW trending tectonically controlled fractures of post cratonic times (Pati et al., 2007, 2008). The Bastar craton also comprises Neoarchaean (~3.5 Ga; U–Pb zircon geochronology of Sarkar et al., 1993; Ghosh, 2004) basement gneissic complex with supracrustal rocks, generally described as the Bengpal and Bailadila Groups and include ~2.5 Ga granitic plutonism (Crookshank, 1963; Ramakrishnan, 1990; Ramchandra et al., 1995; Ramakrishnan and Vaidyanadhan, 2008). The northern boundary of Bastar craton is represented by the Central Indian Tectonic Zone (CITZ; Fig. 1) which is regarded as a major tectonic divide between the northern (Bundelkhand and Aravalli) cratons and southern (Bastar, Singbhum and Dharwar) cratons in the Indian shield (Acharyya, 2003; Zhao et al., 2004). Proterozoic intracontinental sedimentary basins were developed over the basement region of Bundelkhand and Bastar cratons. Mafic dykes constitute major Proterozoic mafic igneous activity in both the cratons and are not seen to cut across the sedimentary basins (Figs. 2 and 3).

The entire Bundelkhand craton is traversed by mafic dykes crossing all Archaean lithologies. The dykes are traced in strike extensions from a few metres up to 20 km and predominantly in a NW–SE direction; ENE–WSW and NE–SW trending dykes also occur but in much smaller numbers. A prominent ENE–WSW dyke is over 50 km long and >20 m wide and passes through the village Mohaba (Fig. 2). The dykes are not metamorphosed or structurally foliated and can be classified as medium to coarse grained dolerites; larger dykes are gabbroic in nature in the central region and fine-grained chilled margins are observed where contact zones are exposed. Typically, dyke samples show ophitic to subophitic textures with occurrence of plagioclase and clinopyroxene phenocrysts/microphenocrysts and are subalkaline tholeiites in composition. Deuteric alteration is common and hydrothermal alteration to chlorite/actinolite is also present. Opaque oxides occur as dispersed grains or skeletal shapes.

Three generations of dykes are proposed based on field dispositions (Basu, 1986) and NW–SE dykes are cut by ENE–WSW and

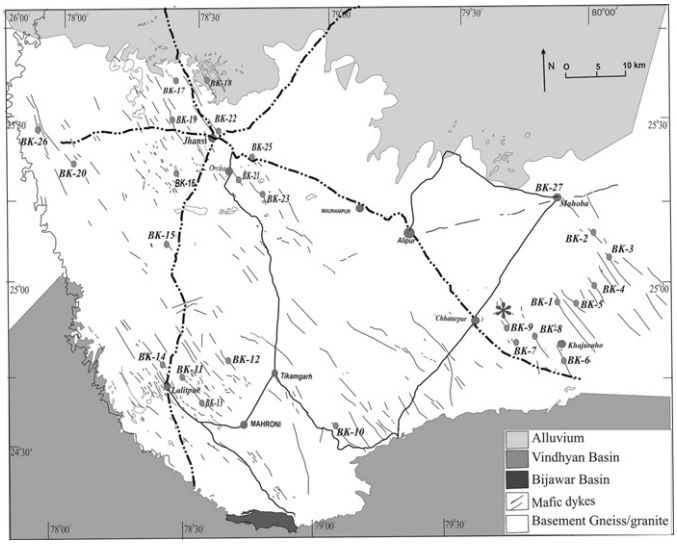


Fig. 2. Geological map of Bundelkhand craton showing mafic dykes compiled from the quadrangle geological maps published by the Geological Survey of India. The palaeomagnetic site locations are shown as dots. Star denotes the dyke dated by U–Pb isotopic dating of zircon fractions by Pradhan et al. (2012). Also marked the site locations of another recent palaeomagnetic study by Pradhan et al. (2012).

NE–SW dykes, although these cross-cutting relationships are not unequivocal and no clear distinctions can be established. Age control on the Bundelkhand mafic dykes is problematic. Earlier K–Ar data loosely constrain ages between ca. 2.1 and 1.5 Ga (Crawford, 1970; Crawford and Compston, 1970). More recent in situ $^{40}Ar/^{39}Ar$ laser ablation data (Rao, 2004; Rao et al., 2005), although without specific sample locations data, show some cluster at ca. 2100 Ma and ca. 2000 Ma. These data include ages as old as 2500 Ma and also much younger ages of ca.700 Ma and suggest that some spot ages are the result of excess/loss of argon. Despite these problems, these data suggest that dyke emplacement is of Palaeoproterozoic age although it is not possible to identify distinct periods of dyke intrusions. Pradhan et al. (2012) obtained a 1985.7 ± 9.4 Ma U–Pb zircon age from well faceted grains of a NW–SE trending dolerite dyke and an 1113 ± 7.4 Ma age from zircon grains/fragments of Mohaba dyke.

In the Bastar craton, the mafic dykes trend predominantly NW–SE to WNW–ESE (Fig. 3) although NE–SW orientations are rarely found. Individual dykes can be traced for several kilometres and are typically about 50 m thick with a few dykes ranging up to 200 m in thickness. Mafic dykes in the Bastar craton include both metamorphosed amphibolites and unmetamorphosed medium to coarse grained dolerites and rarely gabbros (Crookshank, 1963; Ramakrishnan, 1990; Ramchandra et al., 1995; Srivastava et al., 1996). Srivastava and Singh (2003, 2004) described a suite of high-Mg rocks as boninites and include them also as a set of mafic dykes. The dolerite dykes are petrographically characterized by ophitic to subophitic textures and contain plagioclase and clinopyroxene phenocrysts/microphenocrysts. Ramchandra et al. (1995) described the dolerite dykes as predominantly quartz-normative subalkaline tholeiites with subordinate proportions of olivine- and nepheline-normative dykes. Hypersthene is also present in some cases when the dykes are noritic. Low-grade uralitisation is present leading to formation of tremolite and chlorite. Disseminated titanomagnetite and minor pyrrhotite, pyrite and chalcopyrite constitute the opaque mineral assemblage.

There are no clear cross-cutting field relations to establish distinct age groups in the mafic dykes. A U–Pb rutile age of 2100 ± 11 Ma was interpreted as a minimum age of emplacement for a metamorphosed dyke (French, 2007; Srivastava et al., 2011) and French et al. (2008) obtained high precision U–Pb baddeleyite and zircon ages of 1891.1 ± 0.9 and 1883 ± 1.4 Ma for two unmetamorphosed dolerite dykes from southern Bastar craton; one of these dykes is well within the present study area (Fig. 3) while the other dyke is a few kilometres south of this study area. French et al. (2008) obtained a comparable U–Pb baddeleyite age for the Pullivendla sill in the Cuddapah Basin (1885.4 ± 3.1 Ma) in the Dharwar craton of the south Indian shield and suggested that the Bastar-Cuddapah mafic igneous activity constitutes a large igneous province.

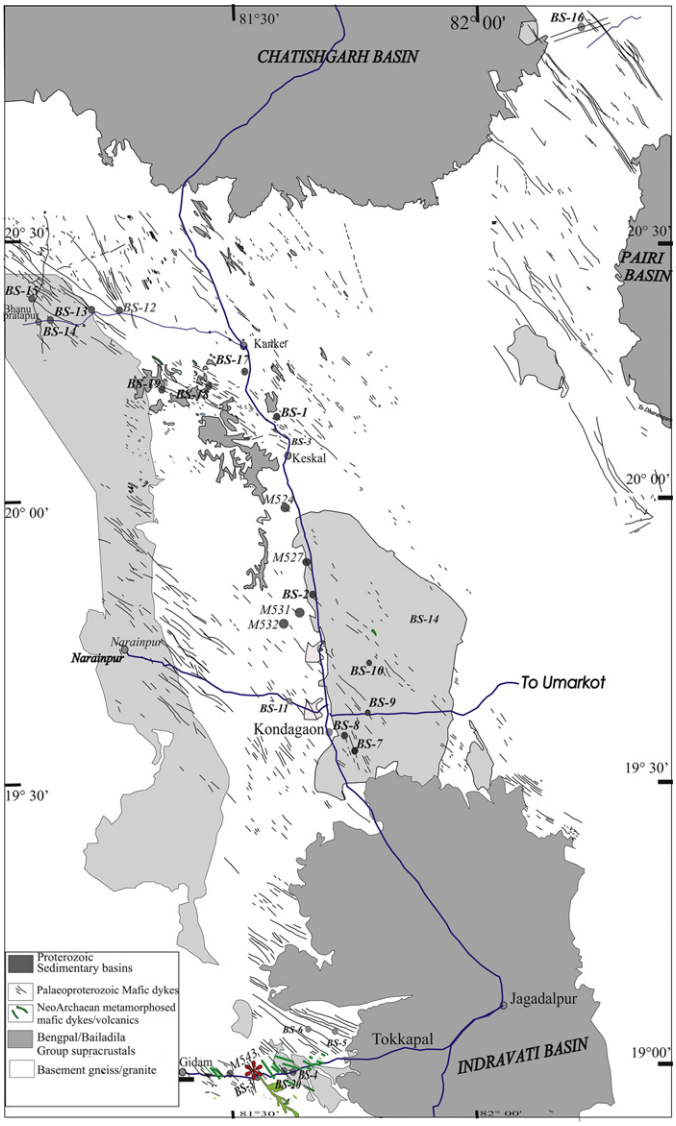


Fig. 3. Generalized geological map of Bastar craton showing the distribution of mafic dykes compiled from the quadrangle and district geological maps published by the Geological Survey of India. The palaeomagnetic site locations are shown as dots. Star denotes the dyke dated by U–Pb isotopic dating of baddeleyite fractions by French et al. (2008). Also marked are the site locations of another recent palaeomagnetic study by Meert et al. (2011).

3. Palaeomagnetism

3.1. Sampling and methods

The whole geographic area of mafic dyke distribution from the Bundelkhand and a major part of the Bastar cratons has been covered for palaeomagnetic study and sampling sites are shown in Figs. 2 and 3. Each site represents a dyke and 10–15 oriented cores

drilled from a dyke using a gasoline-powered portable drill (ASC Scientific D2801; USA) to constitute a site. An in-house fabricated orientation stage and magnetic compass were used for recording core orientations and markings were always made by keeping the magnetic compass about 20 cm above the rock surface to minimize errors due to outcrop magnetization. In total 610 samples from 47 sites (342 samples from 27 dykes in the Bunelkhand and 268 samples from 20 dykes in the Bastar) were available for palaeomagnetic study. The sites BK-7 and BK-27 are respectively the strike extensions of dykes dated by Pradhan et al. (2012) at ca. 1.99 Ga and 1.1 Ga and the BS-3 site is the strike extension of a dyke dated by French et al. (2008) at ca. 1.89 Ga. Cylindrical specimens of 2.54 cm diameter and length were obtained for analysis.

All specimens were subjected to natural remnant magnetism (NRM) measurements and step-wise incremental alternate field (AF) demagnetization was carried out at 2.5 mT steps (1.5 mT in a few samples where intensities are low) up to 100 mT or until stable magnetization vectors were delineated, or the intensities fell below the sensitivity (0.5 mA/m) of the instrument. The experiments were conducted in the palaeomagnetism laboratory of the Centre for Earth Science Studies, Trivandrum. MOLSPIN MINISPIN Spinner Magnetometer (MOLSPIN Ltd; UK) and the AF shielded demagnetizer (MOLSPIN Ltd; UK) were used for measuring magnetizations and for AF demagnetizations respectively. Step-wise progressive incremental thermal demagnetization was also carried out on representative samples using MMTD Thermal Demagnetiser (Magnetic Measurements; UK). Thermal demagnetisations were performed at RT, 50–500 °C in 50 °C steps and in 20 °C steps at 500–600 °C. Standard orthogonal projections of magnetic remanence vectors and stereographic projection methods were employed to delineate component structures and isolate characteristic magnetisations. Linear regression tests using principal component analysis (Kirschvink, 1980) were employed to calculate component directions. Only those linear trajectories of consecutive points with an angular uncertainty of <10° were accepted. Mean site directions were calculated from individual sample ChRMs. Based on overlap of α_{95} circles, the ChRM directions were classified into distinct ChRM groups and a mean direction calculated for each group from the site mean directions corresponding to the group. The mean calculations followed Fisher's statistics. Temperature–susceptibility values during heating and cooling cycles and Isothermal Remanence Acquisition (IRM) were recorded for representative samples using T-χ apparatus (Bartington Ltd; UK) and an IM30 impulse magnetizer (ASC Scientific, USA), respectively.

3.2. Results

The T–χ curves (Figs. 4i–l and 5g, h) are generally reversible during heating–cooling cycles. The IRM experiments show saturation at about 300 mT with very low (<30 mT) coercivity values (Fig. 6). The typical magnetic carrier in the mafic dykes of the Bundelkhand and Bastar cratons was found to be low-Ti titanomagnetite with Curie temperatures in the range of 540–560 °C. The presence of ilmenite and/or pyrrhotite (or chalcopyrite) is also evident in some samples (example, BK-14j) with minor fall of susceptibility between 250 and 300 °C. In sample BS-14h, low temperature titanomagnetite (maghemitisation) possibly related to hydrothermal alteration appears to be present as the susceptibility exhibits minor rise at about 200 °C and the cooling curve is above the heating curve.

Mafic dykes from both Bundelkhand and Bastar cratons are Fe-rich tholeiitic basalts and are strongly magnetized with intensities typically of the order of 1–10 A/m and readily measured on Minispin spinner magnetometer. In the whole collection, ChRM directions of 2 sites from the Bundelkhand (BK-13; BK-27) and 5 sites (BS-3; BS-4; BS-5; BS-16; BS-29) from the Bastar craton show no coherent groupings and are not discussed further. Samples from the remaining sites have scattered NRMs, however, AF demagnetisations with peak fields in the range of 10–20 mT have generally isolated stable directions and achieved within-site internal consistency in a majority (nearly 50–70%) of the samples. Thermal demagnetisation usually subtracts the NRM component without any significant change in direction. In a few samples, specimens subjected to both thermal and AF treatments have different NRM. In these samples, the NRM direction of AF specimens becomes similar to those of thermal specimens after the first 2/3 steps of AF cleaning. However, further AF treatment subtracts the component and isolates the ChRM while the thermal treatment subtracts the same inherent NRM component. Thus the AF demagnetisation experiments are found useful to isolate the ChRM. Typical examples of AF and thermal demagnetisation responses are shown as orthogonal projections in Figs. 4 and 5. The mean site directions fall into four distinct groups in Bundelkhand and three in Bastar. The mean ChRMs of the sites (dykes) from both the cratons are given in Tables 1 and 2 and shown in Figs. 7 and 8. The α_{95} confidence circles of all mean directions within each group show marked overlap. Therefore, we are able to confidently distinguish distinct directions namely northwest (and antipodal southeast) shallow, easterly (and antipodal westerly) shallow and steep upward/downward directions. Northeast (and near antipodal southwest) shallow directions are found in Bundelkhand craton but this is not retrieved from the Bastar dykes.

More than one-third of the Bundelkhand dykes which yielded within-site grouping (nine sites: BK-1, BK-4, BK-5, BK-7, BK-8, BK-19, BK-21, BK-23 and BK-24) constitute the southeast shallow ChRM group (Fig. 7). All these dykes have a NW-SE strike trends (Fig. 1). The α_{95} values of these sites are <16° except in two sites (BK-4 and BK-23) where it is a little higher (<17°). Site BK-9 has also yielded a comparable direction with large α_{95} value of 38°, but the mean value of this site falls within the α_{95} circle of the mean direction of this group (Fig. 7) and this site is therefore included into this group. The calculated mean direction from these 10 sites is well constrained (D = 155° and I = 1°; K = 33; α_{95} = 8.5°). A comparable direction is obtained for ten NW–SE trending dykes (12 sites) in a recent study by Pradhan et al. (2012). Combining the data from these two studies, we derive a grand mean direction of this group of directions of D = 155° and I = −5° (k = 26; α_{95} = 6.2°; N = 22). It is noteworthy that none of the dykes of our collection from Bastar craton yield this direction. However, directions from two dykes (site I527 and I532 reported by Meert et al., 2011) show significant overlap with this group in terms of α_{95} circles and we regard them to belong to this group (Table 2; Fig. 8). The southeasterly shallow directions of Bundelkhand include both polarity (15 dykes of southeast shallow upward and 7 antipodal dykes) magnetizations, which contrast with the dominant antipodal polarity in the south Indian shield. A reversal test (McFadden and McElhinny, 1990) performed between both the bipolar characteristic remanences passes the test with classification B with γ = 9.3°. In the Bastar craton, both dykes included in this group are of single polarity with southeast shallow magnetizations.

Dykes of near-easterly shallow directions are more predominant in the Bastar craton. Six out of fifteen dykes (BS-2; BS-7; BS-11; BS-14; BS-17; BS-18) yielding within-site grouping fall into this group (Fig. 8A; Table 2). Two dykes (BS-6 and BS-10) yield comparable mean directions with high α_{95} values, but the site mean values are within α_{95} circles of other sites in this group (Fig. 8A) and are considered to relate to the group. Using data from the 8 dykes of our study, a mean direction of this group is calculated (D = 106°; I = −4°; k = 21; α_{95} = 12.3°; Table 2). A study carried out recently on dykes in Bastar craton by Meert et al. (2011) derived a coherent site mean directions for three dykes overlapping the near easterly shallow

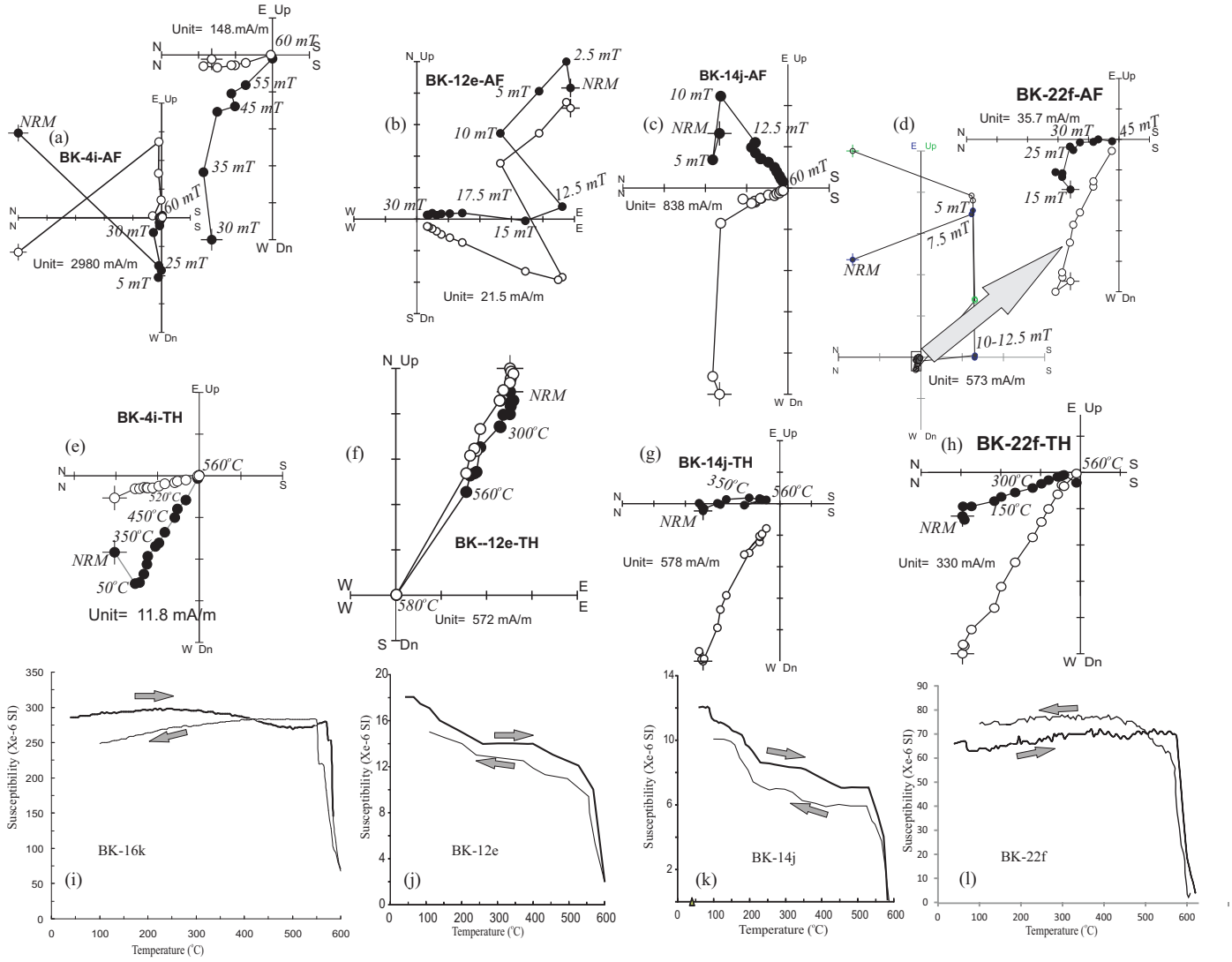


Fig. 4. Orthogonal projections of magnetic remanence from representative samples of Bundelkhand mafic dykes Suffixes "AF" and "TH" denote alternating field and thermal demagnetizations, respectively. Open and closed symbols denote projections onto the horizontal and vertical planes, respectively. (i–l) Thermomagnetic plots depicting susceptibility–temperature variations for the sites.

directions of our study in terms of α_{95} circles. Therefore, combining the data of these three dykes with the data from 8 dykes of our study, a grand mean ChRM direction of this group is calculated for the Bastar craton ($D = 108^\circ$; $I = -0.03^\circ$; $k = 21$; $\alpha_{95} = 10.2^\circ$; Table 2). In the Bundelkhand craton, the near-easterly shallow direction is less predominant. It is recovered from only three dykes (sites BK-12, BK-25 and BK-26; Table 1). The α_{95} values range between 15° and 19°. Occurrence of dykes of this group is established based on good between-site correlations evident in terms of overlapping of α_{95} circles (Fig. 7b). In the case of near easterly shallow magnetizations, the dykes with both polarity magnetizations (7 dykes with near easterly directions and 4 dykes with antipodal directions) are located in the Bastar craton. A reversals test defines a class C with $\gamma = 12.7^\circ$. In Bundelkhand craton, dykes with comparable magnetization are much less abundant and have only unipolar near easterly shallow magnetizations (Fig. 7).

Both in Bundelkhand and Bastar cratons, the steep upward/downward directions constitute a further magnetization groups with statistically distinguishable ChRM directions. The poles estimated from these dykes can further be classified into three subgroups. A steep group designated steep 1 directions is recognized in five dykes in Bastar (BS-1; BS-8; BS-12; BS-13; BS-19) yielding a mean direction of $D = 283.1^\circ$; $I = 84.9^\circ$ ($k = 44$; $\alpha_{95} = 11.7^\circ$) with a corresponding palaeopole at $\lambda = 22.3^\circ$N; $\Phi = 71.4^\circ$E; $A_{95} = 21.6^\circ$ (Fig. 8b and Table 2). A comparable direction is found in 4 Bundelkhand dykes (two sites of this study and two sites of Pradhan et al., 2012; Table 1) and a combined mean is calculated for these four sites ($D = 188.6^\circ$; $I = 69.8^\circ$; $k = 67$; $\alpha_{95} = 11.3^\circ$) yielding a corresponding palaeopole at $\lambda = 10.2^\circ$S; $\Phi = 75.0^\circ$E; $A_{95} = 18.1^\circ$. The steep 1 magnetisations in the Bundelkhand craton are unipolar with downward inclinations, but in the Bastar craton both upward and downward inclinations are recovered (2 upward and 3 downward) and a reversals test defines a C classification with $\gamma = 18^\circ$. Another direction designated as steep 2 direction is found in BK-10 dyke (Bundelkhand; $\lambda = 14^\circ$S; $\Phi = 101^\circ$E; $A_{95} = 26.1^\circ$) and BS-9 dyke (Bastar; $\lambda = 06^\circ$N; $\Phi = 113^\circ$E; $A_{95} = 26.5^\circ$). Both dykes possess upward inclinations. In the Bundelkhand craton, a coherent set of distinct steep directions (steep 4) is recorded in five dykes with the palaeopole situated at intermediate-steep latitudes ($\lambda = 60.4^\circ$N; $\Phi = 45.3^\circ$E; $A_{95} = 9.7^\circ$) in contrast to the equatorial palaeopoles of steep 1 and 2 directions. The steep directions of this group comprise four downward inclinations and one dyke yielded an upward

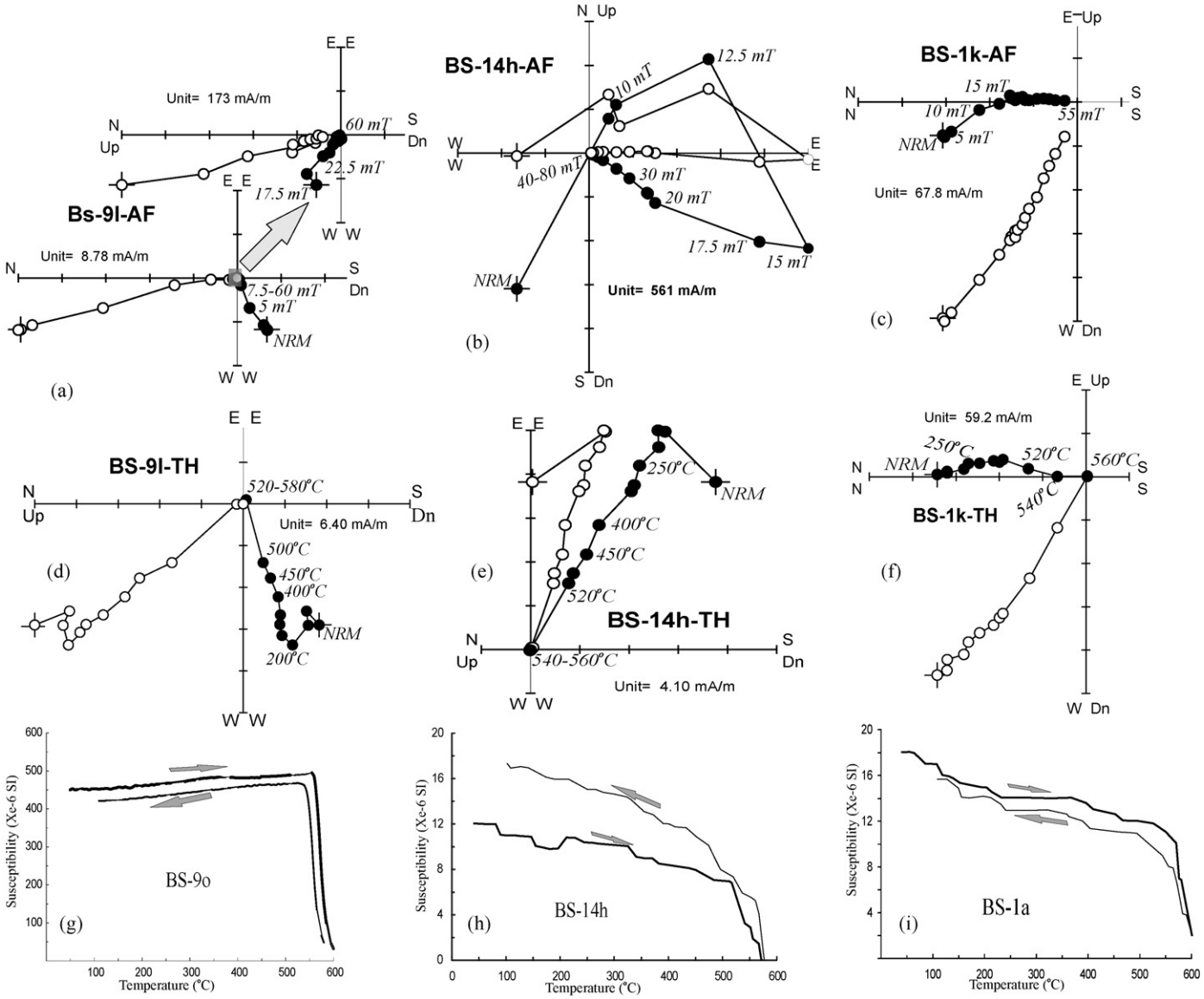


Fig. 5. Orthogonal projections of magnetic remanence from representative samples of the Bastar mafic dykes Suffixes "AF" and "TH" denote alternating field and thermal demagnetizations, respectively. Open and closed symbols denote projections onto the horizontal and vertical planes, respectively. (g–i) Thermomagnetic plots depicting susceptibility-temperature variations for the sites.

inclination representing antipodal direction. This steep direction with an intermediate palaeopole latitude is found in BS-15 site of the Bastar craton ($\lambda = 49°$N; $\Phi = 129°$E; $A_{95} = 15.1°$).

Northeast shallow downward ChRMs ($D = 53.5°$; $I = 15°$; $k = 296$; $\alpha_{95} = 14.5°$) are found in two dykes (BK-11; BK-14) of the Bundelkhand craton (Table 1 and Fig. 7). A direction in close proximity is found in four sites from a large dyke passing through Mohaba village in the study of Pradhan et al. (2012), although our site from the same dyke yielded no reliable direction. The direction from the Mohaba dyke has lower declination ($D = 24.7°$) and near-moderate upward inclination ($I = -37.9°$) and its α_{95} confidence circle does not incorporate the mean direction of any one site of our study (Fig. 7). Furthermore, our results from two dykes (BK-2 and BK-3) yield a similar antipodal direction closer to the remanence in the Mohaba dyke and the α_{95} confidence circles are distinctive from those of BK-11, BK-14 sites. Hence, we regard the Mohaba dyke and probably dykes BK-2 and BK-3 as belonging to another group of directions. Our collection and the study by Meert et al. (2011) from Bastar craton found no comparable direction in that craton.

There is pronounced within-site spread in terms of precision parameters at individual site level compared to between-site precision parameters and occasionally some samples show a reverse magnetization. This indicates different cooling histories for the dykes and remanence acquisition over a protracted cooling period. These data would thus tend to average out the geomagnetic secular variation.

4. Discussion

4.1. Comparison of the results across the Indian shield and ages of magnetisations

There have been some preliminary reports of palaeomagnetic data on the Bundelkhand craton (Rao et al., 2005; Patil et al., 2006), lacking details on statistics or methods and these are not considered further. The present study along with the results of Meert et al. (2011) and Pradhan et al. (2012) constitutes the first major palaeomagnetic study from both Bundelkhand and Bastar

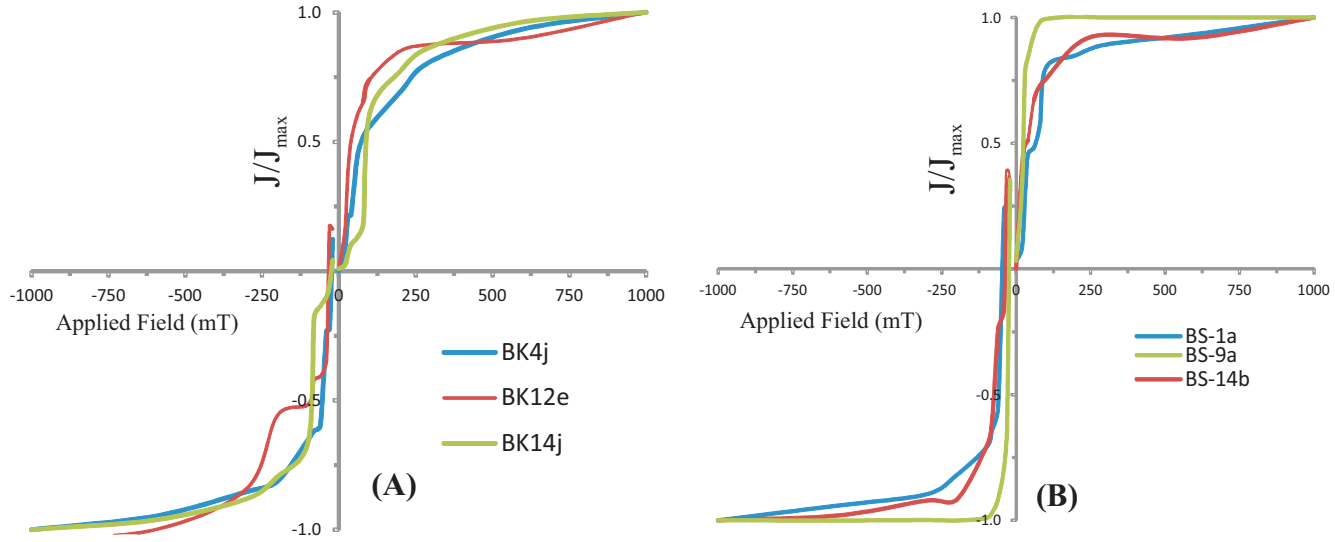


Fig. 6. Examples of Isothermal remanent magnetization acquisition curves during application of forward and back-field for dyke samples from Bundelkhand (A) and Bastar (B) cratons.

Table 1
Palaeomagnetic results from dykes of the Bundelkhand craton.

Site	Trend	Lat	Long	*N*	Ns	*D*	*I*	*k*	α_{95}	λ	*Φ*
Steep ChRM											
Steep 1											
BK6	NW	24.76	79.90	10	6	165	80	24	17.2	06°N	085
BK16	NW	25.38	78.53	16	5	198	69	14	20.6	11°S	68
BK-S1 2.37 Ga sites (including Pradhan et al., 2012)					4	188.6	69.8	67	11.3	10.2°S	75.0 (A_{95} = 18.1°)
Steep 2: BK-S2 2.45 Ga											
BK10	NW	24.74	78.86	11	5	329	−63	20	17.7	14°S	101 (A_{95} = 26.1°)
Steep 4											
BK15	NW	.13	78.45	13	6	165	−66	18	16.2	64°S	235
BK17	NW	25.58	78.61	12	6	335	67	33	11.9	59°N	46
BK18	NW	25.64	78.63	13	6	316	64	103	6.6	50°N	30
BK22	NW	25.50	78.59	12	7	341	58	86	7.3	69°N	33
BK20	NW	25.43	78.09	12	4	345	72	9	33.3	57°N	63
Mean					5	336.0	65.8	129	6.8	60.4°N	45.3 (A_{95} = 9.7°)
BK-SE; 2.0 Ga: SE shallow ChRM											
BK1	NW	24.95	79.91	12	7	160	−20	68	7.4	66°S	136
BK4	NW	25.07	80.16	14	5	151	−14	21	17.0	57°S	142
BK5	NW	24.84	79.97	12	6	160	26	20	15.6	47°S	109
BK7	NW	27.80	79.80	13	5	157	6	50	10.9	52°S	119
BK8	NW	24.90	79.78	16	7	343	−2	25	12.5	59°N	295
BK19	NW	25.46	78.48	11	7	146	3	23	12.8	48°S	134
BK21	NW	25.41	78.65	14	10	147	7	21	10.8	47°S	101
BK23	NW	25.39	78.77	9	4	163	2	32	16.6	59°S	113
BK24	NW	25.37	78.76	6	4	329	25	42	14.3	58°N	332
BK9	NW	24.93	79.69	15	4	150	−4	7	38.0	53°S	136
Mean					10	156	1	33	8.5	56°N	305 (A_{95} = 6. 0)
Pradhan et al. (2012)					12	155.3	−7.8	21	9.6	59°N	313 (A_{95} = 7.0°)
Grand mean (including Pradhan et al., 2012)					22	155	−5	26	6.2	57.5°N	309.0 (A_{95} = 4.7°)
Easterly shallow ChRM											
BK12	NW	24.73	78.54	15	7	104	14	16	15.5	10°S	156
BK25	NW	25.37	78.76	13	5	84	−12	25	15.8	3°N	177
BK26	NW	25.47	78.14	10	5	83	7	18	18.7	8°N	168
Mean					3 sites	90	3	21	27.5	0.4°S	347 (A_{95} = 21.6°)
Northeast shallow ChRMs											
BK11	NW	24.70	78.46	15	3	52	18	38	20.2	38°N	176
BK14	NW	24.74	78.43	15	5	55	12	24	15.6	34°N	178
Mean					2	53.5	15.0	296	14.5	36.0°S	357.0 (A_{95} = 9.4°)
BK2	NW	25.13	80.04	15	5	207	10	8	28.9	50°S	035
BK3	NW	25.07	80.16	15	4	202	−1	25	18.6	58°S	036
Mean					2	205	4.5	90	26.6	54°S	036 (A_{95} = 17.6°)
BK13	NW	24.61	78.61	14	3		No grouping				
BK27	NE	25.28	79.88	10	3		No grouping				

N, number of samples studied from the site; Ns, number of samples yielded stable directions constituting the group; *D* and *I* are declination and inclination (in degrees), respectively, of the characteristic remnant magnetic directions; *k*, precision parameter; α_{95}, radius of the 95% confidence; λ and *Φ* are, respectively, the latitude and longitude of the VGPs calculated.

Table 2
Palaeomagnetic results from dykes of the Bastar craton.

Site	Trend	Lat.	Long.	N	Ns	D	I	k	α_{95}	λ	Φ
Steep ChRM											
Steep 1											
BS1	NW	20.13	81.56	13	5	242	69	22	16.6	0.3°N	49
BS8	NW	19.56	81.71	15	6	138	−83	13	19.3	29°S	251
BS12	NW	19.33	81.24	14	5	247	−87	13	22.2	22°S	267
BS13	NW	20.32	81.20	15	9	216	84	15	13.9	11°N	74
BS19	NW	20.20	81.35	11	8	176	−76	38	9.2	47°S	259
					5	283.1	84.9	44	11.7	22.3°N	71.4 (A_{95} = 21.6°)
Steep 2											
BS9	NW	19.61	81.74	16	5	289	−71	21	17.1	06°N	113 (A_{95} = 26.5°)
Steep 4											
BS15	NW	20.35	81.06	15	5	220	−60	49	11.0	49°S	309 (A_{95} = 15.1°)
East-southeast shallow ChRM											
BS2	NW	19.83	81.64	15	10	118	8	47	7.2	25°N	157
BS7	NW	19.55	81.72	15	6	286	−1	17	16.9	15°N	346
BS11	NW	19.63	81.60	14	9	121	10	22	11.4	27°S	155
BS14	NW	20.31	81.09	17	6	106	14	21	14.9	12°S	159
BS17	NW	20.20	81.50	10	5	101	−19	32	13.8	14°S	177
BS18	NW	20.20	81.46	10	7	99	−29	58	8.0	14°S	183
BS6	NW	19.05	81.62	11	4	283	3	6	39.2	13°N	349
BS10	NW	19.68	81.74	14	4	94	−12	8	35.0	06°S	176
Mean					8	106	−4	21	12.3	9.9°N	348.0 (A_{95} = 12.6°)
Grand mean including 3 sites of Meert et al. (2011)					11	108	0	21	10.2	18.0°N	344.0 (A_{95} = 8.1°)
Northwest shallow											
Meert et al. (2011)					2	137	18	412	39.5	39°N	321 (A_{95} = 28°)
Others											
BS3	NW	18.98	81.56	12			No stable vectors				
BS4	NW	18.99	81.62	16			No grouping				
BS5	NW	19.06	81.65	12			No grouping				
BS16	NW	20.84	82.21	9			No grouping				
BS20	NW	18.97	81.56	10			No grouping				

Explanations are same as in Table 1.

cratons. These studies record dual polarity magnetizations and are representative of large cratonic areas. They also pass reversals tests and the directions of magnetization are not recognized in post Palaeoproterozoic igneous units except for the steep directions. Although steep directions are known from 1092 Ma kimberlites and 830 Ma alkaline dykes in the eastern Dharwar craton (Miller and Hargraves, 1994; Dawson and Hargraves, 1994; Radhakrishna and Joseph, 1996b), strong thermal overprints of either of these ages are not known from the Bundelkhand and Bastar cratons. Moreover, the kimberlites and alkaline dykes record steep directions with corresponding poles at intermediate latitudes in contrast to the steep1 and steep 2 poles distributed in the equatorial region. An apparent comparison is seen as stated earlier between the northeast shallow directions and the directions of the 1113 ± 7.4 Ma Mohaba dyke (Pradhan et al., 2012) or that of the 1073 ± 13.7 Ma Majhgawan kimberlite (Gregory et al., 2006) whose pole is within 15° of Mohaba dyke. However, the 2.2 Ga pole and the Neoproterozoic pole are spatially much removed (>40°). Therefore, these directions are regarded as primary magnetizations acquired during crystallization of the dykes. In addition, although we could not sample for baked contact test, majority of granitic host rock samples collected for baked contact test from the Bundelkhand craton by Pradhan et al. (2012) did not yield consistent directions because of the dominance of multi-domain magnetite. Nonetheless, host rock samples close to one dyke (I925) that yielded SW shallow negative direction) obtained stable directions comparable to the main dyke suggesting a partial positive baked contact test.

In the descriptions of the preceding section, we show that the different groups of ChRMs found in dykes of Bundelkhand are in good agreement with the directions found in dykes of the Bastar craton. Furthermore, these directions remarkably coincide with groups of directions found in the dykes of Dharwar craton and therefore, represent discrete, but synchronous, magmatic responses across the cratonic elements in India. This deduction can best be tested from the dykes with the shallow southeast group of ChRM because the dykes of this group are recorded across the cratons and high precision U–Pb baddeleyite/zircon ages have been reported recently for dykes with this magnetization. The southeast shallow ChRMs with antipodal northwest shallow directions represent a well constrained magnetization in the dykes of Bundelkhand craton (Table 1). Our dyke site BK-7 possessing a ChRM falling into this group is on the strike extension of a dyke (19GS-13; Pradhan et al., 2012) from which well faceted zircon fractions yields a U–Pb isotopic age of 1991 ± 11 Ma. The Pulivendla Traps from the lower Cuddapah Basin in south India yielded U–Pb baddeleyite ages of 1885.4 ± 3.1 Ma (French et al., 2008). Palaeomagnetic data from the traps (D = 299°; I = −6°; α_{95} = 16°; Clark, 1982) are in close agreement with the southwest direction of the Bundelkhand dykes. Comparable directions have been recovered from dykes around Cuddapah Basin in south India (Fig. 1) and are assigned an age of 1.9 Ga (Radhakrishna et al., 2003, submitted for publication). The ChRMs of the dykes classified as Group II directions in south India by Hargraves and Bhalla (1983) are comparable with this 1.9 Ga ChRM. It appears that the dykes of this ChRM are scarce in Bastar craton. In the total population of directions available from mafic dykes from Bastar craton, two dykes with comparable direction (I532 and I527: Meert et al., 2011) represent this group (Table 2). U–Pb dates of baddeleyite/zircon (1891.1 ± 0.9 Ma and 1883.0 ± 1.4 Ma; French et al., 2008) for two dykes in the Bastar craton attribute an age of 1.9 Ga to the dykes with this direction. Considering the close comparison between the characteristic magnetisations of the mafic dykes from these cratons and proximity of ages obtained, the dyke magmatism corresponding to this ChRM is penecontemporaneous in the cratonic elements of the Indian shield between 1.89 and 1.99 Ga.

The near-easterly shallow ChRM recovered both in Bundelkhand and Bastar cratons is also recovered from mafic dykes in the Dharwar craton around the Cuddapah Basin. The dykes with corresponding ChRMs in the Dharwar craton are classified as Group

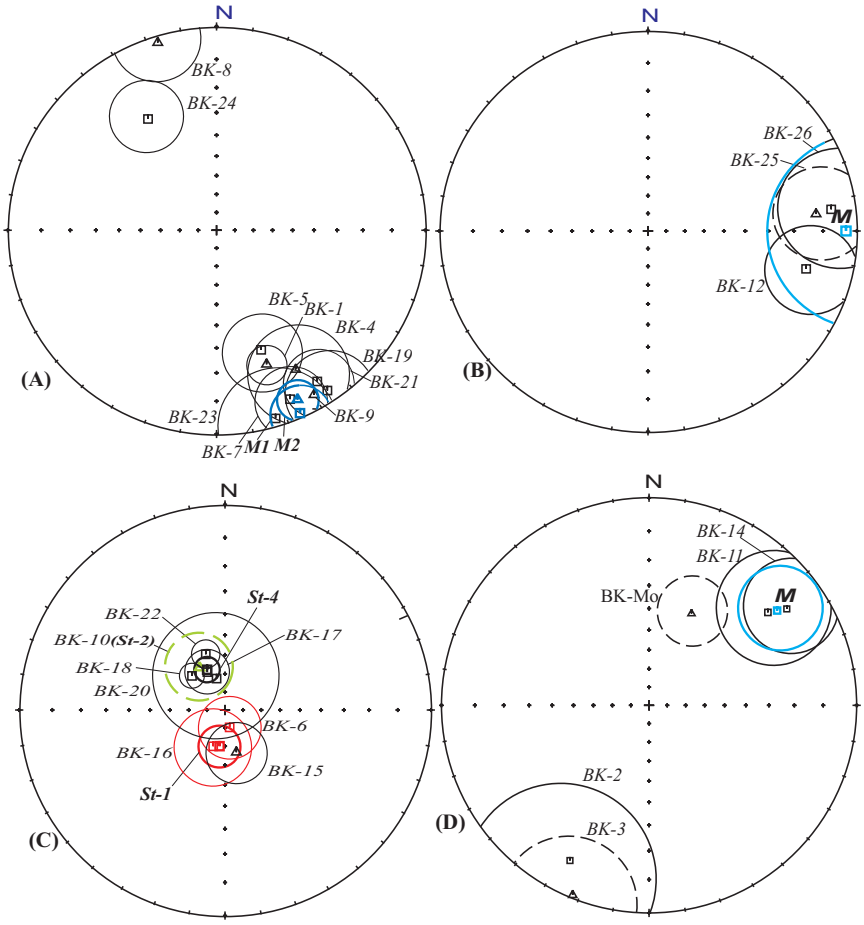


Fig. 7. Equal-area projection of Palaeoproterozoic dykes of Bundelkhand craton showing mean ChRMs with α_{95} confidence circles for the distinct groups of northwest/southeast shallow (A), near easterly/westerly (B), steep (C) and northeast/southwest shallow (D) directions. Circles of continuous and broken lines represent lower and upper hemispheres, respectively. M1 and M2 in (A) are the mean of our study and the mean of combined results of our study and Pradhan et al. (2012). BK-Mo is the Mohaba dyke direction (Pradhan et al., 2012).

I dykes (Hargraves and Bhalla, 1983). More conspicuous is the distinction of polarity of the ChRM of the dykes between the cratons. The near westerly shallow ChRMs are predominant in the south Indian shield (see Fig. 9; Hargraves and Bhalla, 1983; Radhakrishna et al., submitted for publication), whereas the antipodal near easterly shallow ChRMs are dominant in Bundelkhnad and Bastar cratons. Radhakrishna et al. (2007) calculated an age of 2135 ± 82 Ma based on Sm/Nd whole rock-mineral isochron using the isotopic data reported by Pandey et al. (1997) from two dykes, which yielded this group of ChRM directions in the northwest Cuddapah basin. In addition, Radhakrishna et al. (submitted for publication) obtained the easterly shallow ChRM for one of the dykes in the Dharwar craton for which a U–Pb baddeleyite age of 2.1 Ga was reported recently (French and Heaman, 2010). Thus the age of the near-easterly or antipodal westerly shallow ChRM is regarded as 2.1 Ga.

Hargraves and Bhalla (1983) grouped all steep direction dykes into a Group V. However, Radhakrishna et al. (submitted for publication) distinguished three subgroups of steep directions and designated them as steep-1, steep-2 and steep-3 groups. The Steep 1 and steep 2 groups constitute poles situated at near equator with longitudes <85° and >85° respectively and the steep 3 group constitutes the magnetisations yielding poles at intermediate latitudes. No significant overlap between the A95 circles over the mean poles of the distinct groups is apparent. Among the three subgroups of steep directions found in the Bundelkhand and Bastar cratons, two subgroups (steep 1 and steep 2; see Tables 1 and 2) have their equivalents again in all the three cratons. The steep 1 ChRMs are dominant in Bastar and have a moderate presence in Bundelkhand and are profusely distributed in the Dharwar craton. Based on U–Pb baddeleyite ages, the dykes of this group of directions are regarded as intrusions emplaced at ~2.37 Ga (Halls et al., 2007; French and Heaman, 2010; Radhakrishna et al., submitted for publication). The steep 2 direction appears to be less predominant; it was identified in 15 dykes from the Dharwar craton, and is much less in occurrence, but present study found its presence in the Bundelkhand and Bastar cratons (Tables 1 and 2). This direction from the sequence of poles on pole data are older and are assigned an age of ca. 2.4–2.45 Ga (Radhakrishna et al., submitted for publication; also see Fig. 8). It is noted that the steep 2 direction is retrieved from only one dyke in each of the cratons of this study and further study would ascertain the interpretations. The steep 4 direction with intermediate palaeopole latitudes (Tables 1 and 2) is more prevalent in Bundelkhand and rare in Bastar and is not found in Dharwar craton. The pole is much removed spatially from poles of the rest of the groups. No age data are available for this ChRM and it is difficult to assign an age to the steep 4 directions at present.

Integration of palaeomagnetic results (Radhakrishna et al., submitted for publication) and U–Pb baddeleyite ages (French and Heaman, 2010) attributes a 2.2 Ga age for the northeast shallow directions. Mafic dykes of this ChRM group are not found in Bastar and are less prominent in Bundelkhand and have significant presence in the Dharwar craton. Pradhan et al. (2012) found a direction that appears to be similar to it (Fig. 7) from one large dyke

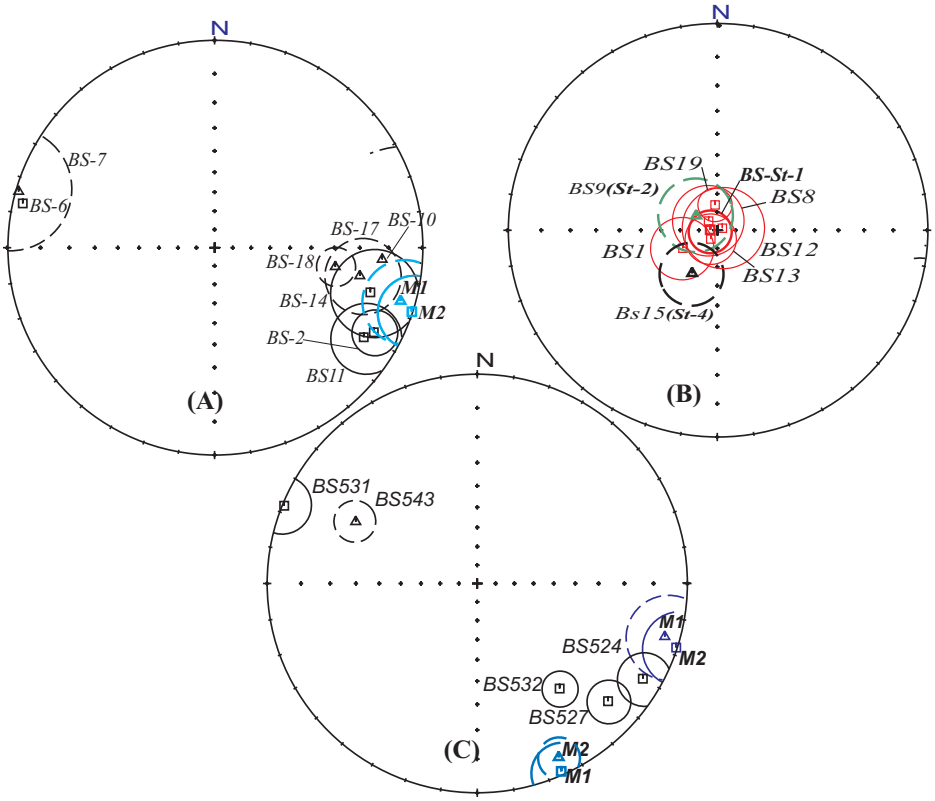


Fig. 8. Equal-area projection of Palaeoproterozoic dykes of Bastar craton showing mean ChRMs with α_{95} confidence circles for the distinct groups of near easterly/westerly (A) and steep (B) directions. In (C) the directions reported by Meert et al. (2011) are projected; two sites (I532 and I527) spatially closer to the southeast shallow confidence circles from Bundelkhand craton (M1 and M2 sky blue). The other three directions have better overlap with the confidence circles of near easterly shallow (or antipodal) directions shown in Neptune blue. Circles of continuous and broken lines represent lower and upper hemispheres, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

passing through the village of Mohaba. However, the Mohaba dyke remanence, as stated earlier, appears to be distinctive from the ca. 2.2 Ga dyke directions and these authors obtained a Neoproterozoic U–Pb zircon age (1113 ± 7.4 Ma) from this dyke. Further studies are required to constrain better these Neoproterozoic directions and the corresponding dyke magmatism.

4.2. Comparisons with other continental blocks and implications for Palaeoproterozoic reconstructions

Several Archaean cratons amalgamated into supercratons or possibly, a supercontinent as a result of accretion or collision process along the 2.0–1.8 Ga orogenic belts (Hoffman, 1988; Gower et al., 1990, 1992; Krapez, 1999; Rogers, 1996; Condie, 2002; Zhao et al., 2002, 2004). Bleeker (2003) suggested that multiple break-up and dispersal events may mean that cratons which are now widely dispersed were once close neighbours. This raises the question as to when the different cratons (Bundelkhand, Bastar, Dharwar, Singbhum) amalgamated to form the Indian shield as a coherent unit and whether the data from discrete cratons are applicable to the whole of the Indian shield. This question is specifically pertinent particularly because the Central Indian Tectonic Zone (CITZ) has been proposed to be a collision zone between the northern and southern cratons in India and suggested to be of the Grenville equivalent as an extension of the Alfred Fraser province in Australia and therefore relates to a supposed Rodinia amalgamation, or an earlier suture of Mesoproterozoic or of late Palaeoproterozoic (1.7–1.9 Ga; see Mishra et al., 2000; Zhao et al., 2002; Acharyya, 2003; Bhowmik et al., 2005; Bhowmik, 2006; Basu Sarbadhikari and Bhowkim, 2008). However, the characteristic magnetisations obtained for the period between 2.4 and 1.9 Ga in this study on dykes from both the Bundelkhand and Bastar cratons (Tables 1 and 2) strongly correlate with the directions retrieved from dykes of comparable age in the Dharwar craton (Table 3). Accordingly, an apparent polar wander path has been constructed based on good quality palaeomagnetic data from the Indian shield (Fig. 9 and Radhakrishna et al., submitted for publication). Our results from Bundelkhand and Bastar are in good agreement with this pole path and show significant overlap with poles from dykes of comparable ages in the Dharwar craton. Thus, amalgamation and collision along the CITZ must have taken place prior to the 2.4 Ga and probably by ~2.5 Ga, when the different cratons in the Indian shield were amalgamated to become near neighbours as suggested by Stein et al. (2004), Radhakrishna et al. (2009) and Piper (2010). A similar proposal has been put forward by Meert et al. (2011) and Pradhan et al. (2012), but their correlations were based only on one palaeomagnetic result from Dharwar craton and our present interpretations substantiate the conclusion based on a larger number of palaeomagnetic results (Radhakrishna et al., submitted for publication).

It follows from the above discussion that the Bundelkhand and Bastar cratons were contiguous with the Dharwar craton in the Palaeoproterozoic and the combined data are useful for tracing the Indian shield in global continental reconstructions. In a complementary paper, we present more detailed global correlations tracing the Indian shield within global continental reconstructions based on the large palaeomagnetic database from the Dharwar craton (Radhakrishna et al., submitted for publication). However significant aspects can be discussed here using the present results from Bundelkhand and Bastar cratons.

Based primarily on the geological connections and correlations of U–Pb geochronology between globally distributed cratonic nuclei and the presence of the Paleo-Mesoproterozoic orogenic

Table 3
Summary of paleomagnetic poles from South Indian shield used for drawing APWP and comparisons (details in Radhakrishna et al., submitted for publication).

Sl. no.	Direction (age)	*N*	Lat. (°N)	Long. (°E)	α_{95}	Region
c. 2.4–2.45 Ga						
1	Steep 2 (2.45 Ga)	13	14.5	109.7	14.5	ACB
2	Steep 2 (2.45 Ga)	21	17.9	106.7	9.8	WSI
c. 2.37 Ga						
3	Steep 1 (c. 2.37 Ga)	22	−5.8	55.7	8.3	ACB
4	Steep 1 (c. 2.37 Ga)	8	21.7	49.6	5.5	SGT
5	Steep 1 (c. 2.37 Ga)	65	7.2	56.8	4.7	WSI
c. 2.22 Ga						
6	NE shallow (2.22 Ga)	21	−41.5	0.3	7.0	ACB
7	NE shallow (2.22 Ga)	36	−41.4	7.3	5.1	WSI
c. 2.18 Ga						
8	East shallow (2.18 Ga)	18	3.1	341.8	6.2	ACB
9	East shallow (2.18 Ga)	24	4.7	342.4	5.3	WSI
c. 1.99–1.89 Ga						
10	NW shallow (1.99 Ga)	15	50.5	331.4	6.4	ACB
11	NW shallow (1.99 Ga)	24	49.2	332.9	4.8	WSI
1.86 Ga						
12	NNW shallow (1.86 Ga)	13	69.3	288.3	3.4	ACB
13	NNW shallow (1.86 Ga)	5	82.6	259.2	10.3	SGT
14	NNW shallow (1.86 Ga)	26	75.5	280.9	3.7	WSI

N, number of dykes; WSI, Whole South Indian Shield; ACB, around Cuddapah Basin; SGT, Southern Granulite Terrain.

belts (2.1–1.8 Ga), a number of NeoArchaean-Palaeoproterozoic supercontinental assemblages have been suggested with variable positions of the Indian shield proposed. Bleeker (2003) proposed the existence of three "supercratons" which broke apart in the Palaeoproterozoic and spawned away to become cratonic nuclei for subsequent crustal growth even far away from their ancestral fragments. In this proposition, Indian cratonic blocks, particularly the Dharwar craton, are shown to be in close proximity to the Slave craton, in the company of Zimbabwe and Wyoming, as a part of the supercraton called "Sclavia". Zhao et al. (2004) depicted a different

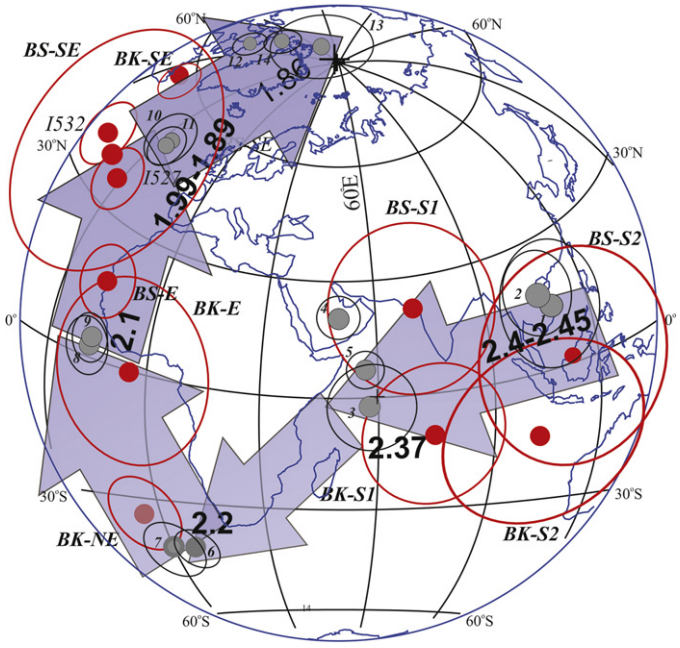


Fig. 9. Mean palaeomagnetic poles with α_{95} confidence circles from mafic dykes of the Bundelkhand and Bastar cratons (red) are plotted over the Palaeoproterozoic path constructed using the data from mafic dyke intrusions in the Dharwar craton, south India (Radhakrishna et al., submitted for publication). The palaeomagnetic pole data of Dharwar dykes along with A_{95} confidence circles and age are also shown (black) for comparison. The Bundelkhand and Bastar pole data are from Tables 1 and 2. The Dharwar pole data labels correspond to the numbers listed in Table 3. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

continental configuration with a close link between north China and India. Similarly Hou et al. (2008) depicted the Indian shield in close contact with north China and Laurentia. However, testing these models using the present palaeomagnetic results from our study and comparable results from other cratons suggests some major differences. The palaeomagnetic ChRMs yield near equatorial palaeolatitudes for the Indian shield for the period 2.2–1.8 Ga and low intermediate latitudes at about 2.37–2.45 Ga (Fig. 10). Although a ca. 2.4 Ga record is not available for the Kaapvaal craton, Laurentia occupied a near-opposite scenario being located in equatorial latitudes at ~2.4 Ga, and both the Laurentia and the Kaapvaal cratons had migrated towards low intermediate latitudes by about 2.2 Ga. In particular, the palaeolatitude from the 2.03 Ga Lac de Gras diabase dykes, the only reliable Palaeoproterozoic palaeomagnetic record from Slave craton (Buchan et al., 2009) is also located away from the 2.1 Ga and 1.99 Ga position estimated for Indian shield in this study suggesting that the Indian shield was not a closest neighbour of the Slave craton. The suggestion is in broad agreement with the recent palaeomagnetic interpretations by Piper (2010). Similarly, in the light of the palaeomagnetic correlations described above suggesting that orogenic activity along the CITZ took place prior to 2.1 Ga and probably ~2.5 Ga, the India–north China connection is probably not valid. Mutually complementary Palaeoproterozoic palaeomagnetic record is not available at present from north China block for robust comparisons. However, a north China block-India linkage is proposed based on palaeomagnetic results from the 1780–1760 Ga dykes (Piper et al., 2011) and well dated Xiong'er Group (Zhang et al., 2012) in north China. Piper et al. (2011) chose to use palaeopole from 1.86 Ga dykes in south India while Zhang et al. (2012) preferred a 1798 ± 120 Ma pole from the Gwalior traps for their comparisons with Indian data. It may be noted that the ages of palaeopoles from India used for comparisons are either older at least by 50 Ma (Piper et al., 2011), or have larger uncertainties as seen in the case of Gwalior traps (Zhang et al., 2012). Furthermore, the early Palaeoproterozoic dykes which are pervasive in the Indian shield are not known in north China. Therefore the proposed north China–India linkage can be at present regarded as tentative and remains to be confirmed. However, it is interesting that the Indian shield and the Yilgarn craton of Australia share nearly identical latitudinal motions between 2.4 and 1.8 Ga (Fig. 10A). Rifting in Palaeoproterozoic as evidenced by the widespread mafic igneous activity and concomitant development of sedimentary basins in western Australia and India

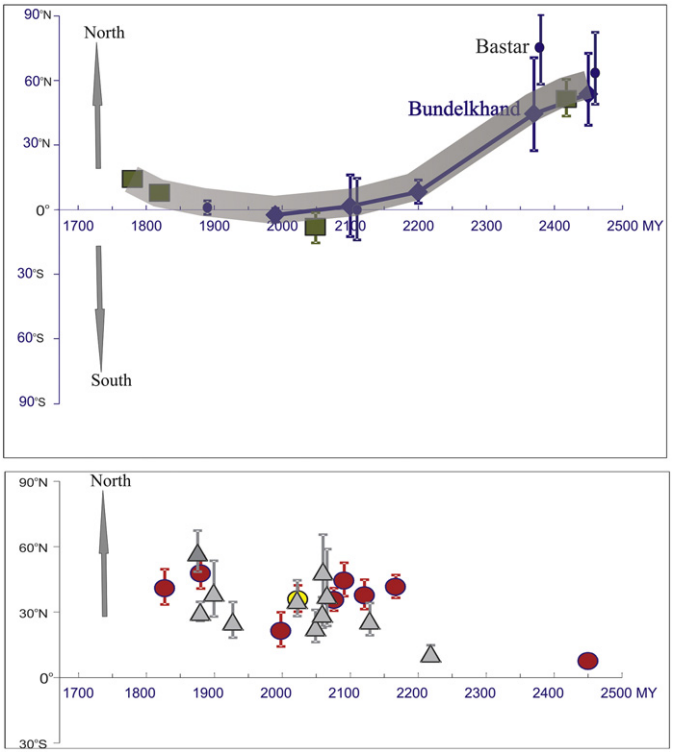


Fig. 10. Palaeolatitude estimates along with error bars of α_{95} confidence limits from (A) the Palaeoproterozoic dykes of Bundelkhand and Bastar cratons (blue) compared with the palaeolatitude data from western Australia (green). In (B) the palaeolatitude estimates for the same period from Kaapvaal (grey) and Laurentia (red) are also plotted for comparison. Palaeolatitude estimates from 2.03 Ga Lac de Gras diabase dykes in the Slave craton is marked in yellow. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)
Data sources are as follows: Australia: 2.41–2.42 Ga Widgiemooltha dykes; Evans (1968) and Nemchin and Pidgeon (1998); 2050 ± 6 Liberia M: Onstott and Dorbor (1987); 1.82 Ga Plum Tree Volcanics: Idnurm and Giddings (1988); 1786 ± 14 Elgee formation: Li (2000); Laurentia: 2446–2473 Matachewan dykes: Bates and Halls (1990); 1998 ± 2 Minto dykes: Buchan et al. (1998); 2167 ± 2 Biscotasing dykes: Buchan et al. (1993); 2121 + 14/−7 Ma Marathon dykes: Buchan et al. (1996); 2091 ± 2 Ma Molson dykes A comp.: Halls and Heaman (2000); 2076 + 5/−4 Ma Fort Frances dykes: Halls (1986); 2027 ± 4 Ma Lac de Gras diabase dykes: Buchan et al. (2009); Kaapvaal: 2222 ± 13 Ma Ongeluk lavas: Evans et al. (1997); 2.15 Ga Southeastern area, N-component: Lubnina et al. (2010); 2130 ± 92 Ma Gamara/Mapedifm: Evans et al. (2002); 2061 ± 274 Ma Bushveld upper complex: Hattingh (1989, 1999); 2050 ± 12 Ma Bushveld complex: Hattingh (1986); 2023 ± 4 Ma Vredefort mean: Salminen et al. (2009); 1.90 Ga Black Hills (NE-trending) dolerite dykes: Lubnina et al. (2010); 1875.5 ± 3.5 Ma Post-Waterberg dolerite: Hanson et al. (2004); 1877 ± 2.2 Ma Mashonaland Sills: Söderlund et al. (2010); 1928 ± 4 Ma Hartley lava: Evans et al. (2002); 2060.5 ± 0.6 Ma Phalaborwa: Evans et al. (2002); 2060.5 ± 0.6 Ma Palabora: Morgan and Briden (1981); 2067 ± 83 Ma Orange River volcanics: Piper (1975).

also provide a case for postulating proximity of these two blocks in the Palaeoproterozoic times (Hoatson et al., 2006; Ramakrishnan and Vaidyanadhan, 2008; Piper, 2010). Thus, the palaeomagnetic data appear to constrain the Yilgarn craton of Western Australia in close proximity to the Indian shield during Palaeoproterozoic. We note that a similar proposition was suggested earlier by Halls et al. (2007). Future palaeomagnetic and geological correlations should refine the above arguments and supercontinental dispersal/amalgamation processes deep in time in the Earth's history.

5. Conclusions

The Bundelkhand and Bastar cratons in the Indian shield are among the oldest cratons of the Archaean and, like many other Precambrian cratons, are traversed by a large number of mafic dykes of Proterozoic age. Palaeomagnetic results of this study identify discrete groups of mafic dykes with distinct ChRMs and provide significant one-to-one correlations with discrete mafic dyke emplacements in the Dharwar craton in south India. Based on integration of available U–Pb geochronology and palaeomagnetic correlations across the cratonic elements in the Indian shield, the mafic dyke emplacements and the discrete ChRMs were assigned ages of ca. 2.45, 2.37, 2.2, 2.1, 1.99 and 1.89 Ga. Strong correlations between the ChRMs and palaeopole data across the cratons within the Indian shield suggest that the cratonic elements within the Indian shield had become close neighbours by about 2.5 Ga. As a corollary, the Central Indian Tectonic Zone, which has been variably interpreted as a 1.9 Ga or Mesoproterozoic collision zone or a result of Rodinia amalgamation in the Neoproterozoic, has been shown to have allegiance to tectonic activity near the time of Archaean/Proterozoic boundary. The palaeolatitudes derived from the palaeomagnetic results of this study suggest that the Indian shield was positioned at higher latitudes during the interval 2.37–2.45 Ga and in an equatorial position during the interval 2.2–1.8 Ga. In contrast, the palaeolatitude estimates of high quality data from Laurentia and Kaapvaal cratons are distinctive. Laurentia was situated near the equator at ca. 2.4 Ga and both Laurentia and Kaapvaal were situated at low-intermediate latitudes during the interval 2.2–1.8 Ga. This comparison is interpreted to mean that the Laurentia and Kaapvaal cratons were not in close proximity as neighbouring crustal blocks to the Indian shield. Interestingly, the Yilgarn craton of Western Australia exhibits a similar latitudinal motion during the interval 2.4–1.8 Ga and is therefore suggested to have been a close neighbour to the Indian shield in the Palaeoproterozoic continental assembly. The tectonic constraints along the CITZ do not support a north China–India linkage in the Palaeoproterozoic. Further investigations from India and other cratons would place better constraints on the position of Indian shield in a supercontinent assembly in the early history of the Earth.

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References

Acharyya, S.K., 2003. The nature of the Mesoproterozoic Central Indian Tectonic Zone with exhumed and reworked older granites. Gondwana Research 6, 197–214.
Anderson, J.L., 1982. Hot-spots, polar wander, Mesozoic convection and the Geoid. Nature 297, 205–208.
Basu, A.K., 1986. Geology of parts of Bundelkhand and Granite Massif, Central India. Geological Survey of India Records 117, 61–124.
Basu Sarbadhikari, A., Bhowkim, S.K., 2008. Constraining the metamorphic evolution of a cryptic hot Mesoproterozoic orogen in the Central Indian Tectonic Zone, using P–T pseudo section modeling of mafic intrusions and host reworked granulites. Precambrian Research 162, 128–149.
Bates, M.P., Halls, H.C., 1990. Regional variation in magnetic polarity of the Matachewan dyke swarm related to the Kapuskasing structural zone, Ontario. Canadian Journal of Earth Sciences 27, 200–211.
Bhowmik, S.K., 2006. Ultra high temperature-metamorphism and its significance in the Central Indian Tectonic Zone. Lithos 92, 484–505.
Bhowmik, S.K., Sarbadhikari, A.B., Spiering, B., Raith, M.M., 2005. Mesozoic reworking of Palaeoproterozoic ultra high temperature granulites in the central Indian tectonic zone and its implications. Journal of Petrology 46, 1085–1119.
Bleeker, W., 2003. The Archean record: a puzzle in ca. 35 pieces. Lithos 71 (2–4), 99–134, http://dx.doi.org/10.1016/j.lithos.2003.07.003.

Buchan, K.L., Mortensen, J.K., Card, K.D., 1993. Northeast trending Early Proterozoic dykes of the southern Superior Province: multiple episodes of emplacement recognized from integrated paleomagnetism and U–Pb geochronology. Canadian Journal of Earth Sciences 30, 1286–1296, http://dx.doi.org/10.1139/e93-110.

Buchan, K.L., Halls, H.C., Mortensen, J.K., 1996. Paleomagnetism, U–Pb geochronology, and geochemistry of Marathon dykes, Superior Province, and comparison with the Fort Frances swarm. Canadian Journal of Earth Sciences 33 (12), 1583–1595, http://dx.doi.org/10.1139/e96-120.

Buchan, K.L., Mortensen, J.K., Card, K.D., Percival, J.A., 1998. Paleomagnetism and U–Pb geochronology of diabase dyke swarms of Minto block, Superior Province, Quebec, Canada. Canadian Journal of Earth Sciences 35 (9), 1054–1069, http://dx.doi.org/10.1139/cjes-35-9-1054.

Buchan, K.L., LeCheminant, A.N., van Breemen, O., 2009. Paleomagnetism and U–Pb geochronology of the Lac de Gras diabase dyke swarm, Slave Province, Canada: implications for relative drift of Slave and Superior provinces in the Paleoproterozoic. Canadian Journal of Earth Sciences 46, 361–379.

Clark, D.A., 1982. Preliminary paleomagnetic results from the Cuddapah traps of Andhra Pradesh. In: Monograph-2, on Evolution of the intracratonic Cuddapah Basin. HPG, Hyde, India, pp. 47–51.

Condie, K.C., 1998. Episodic continental growth models: a mantle avalanche connection? Earth and Planetary Science Letters 163, 97–108.

Condie, K.C., 2002. Breakup of a Paleopreoterozoic supercontinent. Gondwana Research 5, 41–43.

Crawford, A.R., 1970. The Precambrian geochronology of Rajasthan and Bundelkhand, northern India. Canadian Journal of Earth Sciences 7, 91–110.

Crawford, A.R., Compston, W., 1970. The age of Vindhyan system of peninsular India. Quarterly Journal of Geological Society of London 125, 351–371.

Crookshank, H., 1963. Geology of southern Bastar and Jeypore from 351 the from the Bailadila range to the Eastern Ghats. Memoirs of the Geological Survey of India 87, 149pp.

Dawson, E.M., Hargraves, R.B., 1994. Palaeomagnetism of Precambrian dyke swarms in the Harohalli area, south of Bangalore, India. Precambrian Research 96, 157–167.

Evans, M.E., 1968. Magnetization of dikes: a study of the paleomagnetism of the Widgiemooltha Dike Suite, Western Australia. Journal of Geophysical Research 73, 3261–3270.

Evans, D.A.D., Mitchell, R.N., 2011. Assembly and breakup of the core of Paleoproterozoic–Mesoproterozoic supercontinent Nuna. Geological Society of America Bulletin 39, 443–446, http://dx.doi.org/10.1130/G31654.1.

Evans, D.A.D., Beukes, N.J., Kirschvink, J.L., 1997. Low-latitude glaciation in the Palaeoproterozoic era. Nature 386, 262–266.

Evans, D.A.D., Beukes, N.J., Kirschvink, J.L., 2002. Paleomagnetism of a lateritic paleoweathering horizon and overlying Paleoproterozoic red beds from South Africa: implications for the Kaapvaal apparent polar wander path and a confirmation of atmospheric oxygen enrichment. Journal of Geophysical Research 107, http://dx.doi.org/10.1029/2001JB000432 (No. BN12, 2326).

French, J.E., 2007. U–Pb dating of the Paleoproterozoic mafic dyke swarms of the South Indian Shield: implications for paleocontinental reconstructions and identifying ancient mantle plume events. Ph.D. Thesis, University of Alberta, Edmonton, Alberta, 373pp.

French, J.E., Heaman, L.M., 2010. Precise U–Pb dating of Paleoproterozoic mafic dyke swarms of the Dharwar craton, India: implications for the existence of the Neoarchean supercraton Sclavia. Precambrian Research 183, 416–441.

French, J.E., Heaman, L.M., Chacko, T., Srivastava, R.K., 2008. 1891–1883 Ma southern Bastar–Cuddapah mafic igneous events, India: a newly recognized large igneous province. Precambrian Research 160, 308–322.

Ghosh, J.G., 2004. 3.56 Ga tonalite in the central part of the Bastar craton. Journal of Asian Earth Sciences 23, 359–364.

Gower, C.F., Ryan, A.B., Rivers, T., 1990. Mid-Proterozoic Laurentia-Baltica: an overview of its geological evolution and summary of the contributions by this volume. In: Gower, C.F., Rivers, T., Ryan, B. (Eds.), Mid-Proterozoic Laurentia-Baltica. Geological Association of Canada Special Paper 38, pp. 1–20.

Gower, C.F., Schärer, U., Heaman, L.M., 1992. The Labradorian Orogeny in the Grenville Province, eastern Labrador. Canadian Journal of Earth Sciences 29, 1944–1957.

Gregory, L.C., Meert, J.G., Pradhan, V., Pandit, M.K., Tamrat, E., Malone, S.J., 2006. A paleomagnetic and geochronologic study of the Majhgawan Kimberlite, India: implications for the age of the Vindhyan Super Group. Precambrian Research 149, 65–75.

Halls, H.C., 1986. Paleomagnetism, structure, and longitudinal correlation of Middle Precambrian dykes from northwestern Ontario and Minnesota. Canadian Journal of Earth Sciences 23, 142–157.

Halls, H.C., Heaman, L.M., 2000. The paleomagnetic significance of new U–Pb age data from the Molson dyke swarm, Cauchon Lake area, Manitoba. Canadian Journal of Earth Sciences 37 (6), 957–966, http://dx.doi.org/10.1139/cjes-37-6-957.

Halls, H.C., Kumar, A., Srinivasan, R., Hamilton, M.A., 2007. Paleomagnetism and U-Pbgeochronology of easterly trending dykes in the Dharwar craton, India: feldspar clouding, radiating dyke swarms and the position of India at 2.37 Ga. Precambrian Research 155, 47–68.

Hanson, R.E., Gose, W.A., Crowley, J.L., Ramezani, J., Bowring, S.A., Bullen, D.S., Hall, R.P., Pancake, J.A., Mukwakeami, J., 2004. Paleoproterozoic intraplate magmatism and basin development on the Kaapvaal Craton: age, paleomagnetism and geochemistry of <1.93 to <1.87 Gapost-Waterberg dolerites. South African Journal of Geology 107, 233–254.

Hargraves, R.B., Bhalla, M.S., 1983. Precambrian palaeomagnetism in India through 1982: a review. In: Naqvi, S.M., Rogers, J.J.W. (Eds.), Precambrian of South India, vol. 4. Geol. Soc. India Mem., pp. 491–524.

Hattingh, P.J., 1986. The paleomagnetism of the Merensky Reef footwall rocks of the Bushveld Complex. Transactions of the Geological Society of South Africa 89, 1–8.

Hattingh, P.J., 1989. The paleomagnetism of the upper zone of Bushveld Complex. Tectonophysics 165, 131–142.

Hattingh, P.J., 1999. Palaeomagnetism and the Bushveld Complex in Southern Africa: some enigmas. In: Radhakrishna, T., Piper, J.D.A. (Eds.), The Indian Subcontinent and Gondwana: A Palaeomagnetic and Rock Magnetic Perspective, 44. Memoir-Geol. Soc. India, pp. 1–7.

Hoatson, D.M., Jaireth, S., Jaques, A.L., Huleatt, M.B., 2006. A Synthesis of Australian Proterozoic Mafic-ultramafic Magmatic Events: Part 1: Western Australia. 1:3,500,000 scale map. Geoscience Australia, Canberra.

Hoffman, P.F., 1988. United plates of America, the birth of a craton. Annual Review of Earth and Planetary Science 16, 543–603.

Hou, G., Santosh, M., Qian, X., Lister, G.S., Li, J., 2008. Configuration of the Late Paleoproterozoic supercontinent Columbia: Insights from radiating mafic dyke swarms. Gondwana Research 14, 395–409.

Idnurm, M., Giddings, J.W., 1988. Australian Precambrian polar wander: a review. Precambrian Research 40/41, 61–88.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophysical Journal. Royal Astronomical Society 62, 699–718.

Krapez, B., 1999. Stratigraphic record of an Atlantic-type global tectonic cycle in the Palaeoproterozoic Ashburton Province of Western Australia. Australian Journal of Earth Sciences 46, 71–87.

Kusky, T.M., Li, J.H., Santosh, M., 2007. The Paleoproterozoic North Hebei Orogen: North China craton's collisional suture with the Columbia supercontinent. Gondwana Research 12 (1–2), 4–28.

Li, Z.X., 2000. Palaeomagnetic evidence for unification of the North and West Australian cratons by ca.1.7 Ga: new results from the Kimberley Basin of northwestern Australia. Geophysical Journal International 142, 173–180.

Lubnina, N.V., Ernst, R., Klausen, M., Söderlund, U., 2010. Paleomagnetic study of NeoArchean-Paleoproterozoic dykes in the Kaapvaal craton. Precambrian Research 183, 523–552.

McFadden, P.L., McElhinny, M.W., 1990. Classification of the reversal test in paleomagnetism. Geophysical Journal International 103, 725–729.

McMenamin, M.A.S., MacMenamin, D.L.S., 1990. The Emergence of Animals: The Cambrian Breakthrough. Columbia Univ. Press, New York.

Meert, J.G., 2002. Paleomagnetic evidence for a Paleo-Mesoproterozoic supercontinent Columbia. Gondwana Research 5, 207–215, http://dx.doi.org/10.1016/S1342-937X(05)70904-7.

Meert, J.G., Pandit, M.K., Pradhan, V.R., 2011. Preliminary report on the paleomagnetism of 1.88 Ga dykes from the Bastar and Dharwar cratons, Peninsular India. Gondwana Research 20, 335–343.

Miller, K.C., Hargraves, R.B., 1994. Paleomagnetism of some Indian kimberlites and Lamproites. Precambrian Research 69, 259–267.

Mishra, D.C., Singh, B., Tiwari, V.M., Gupta, S.B., Rao, M.B.S.V., 2000. Two cases of continental collisions and related tectonics during the Proterozoic period in India – insights from gravity modeling constrained by seismic and magnetotelluric studies. Precambrian Research 99, 149–169.

Morgan, G.E., Briden, J.C., 1981. Aspects of Precambrian palaeomagnetism, with new data from the Limpopo Mobile Belt and Kaapvaal craton in Southern Africa. Journal Physics of the Earth and Planetary Interiors 24, 142–168.

Nemchin, A.A., Pidgeon, R.T., 1998. Precise conventional and SHRIMP baddeleyite U–Pb age for the Binneringe Dyke, near Narrogin, Western Australia. Australian Journal of Earth Sciences 45, 637–675.

Onstott, T.C., Dorbor, J., 1987. $^{40}Ar/^{39}Ar$ and paleomagnetic results from Liberia and the Precambrian APW database for the West African Shield. Journal of African Earth Sciences 6, 537–552.

Pandey, B.K., Gupta, J.N., Sarma, K.J., Sastry, C.A., 1997. Sm–Nd, Pb–Pb and Rb–Sr geochronology and petrogenesis of the mafic dyke swarm of Mahabubnagar, South India: implications for paleoproterozoic crustal evolution of the Eastern Dharwar Craton. Precambrian Research 84, 181–196.

Pati, J.K., Patel, S.C., Pruseth, K.L., Malviya, V.P., Arima, M., Raju, S., Pati, P., Prakash, K., 2007. Geochemistry of giant quartz veins from the Bundelkhand craton, Central India and its implications. Journal of Earth System Science 116, 510–697.

Pati, J.K., Raju, S., Malviya, V.P., Bhushan, R., Prakash, K., Patel, S.C., 2008. Mafic Dykes of Bundelkhand Craton, Central India: Field, Petrological and Geochemical Characteristics. Narosa Publishing House Pvt. Ltd., New Delhi, pp. 547–569.

Patil, S.K., Nishad, R.K., Seena, M.S., Singh, V.K., 2006. Palaeomagnetic, rock magnetic and low field AMS investigations on the Proterozoic dykes of Bundelkhand Craton: constraints on their age and mode of emplacement mechanism. Nat. Sem. on Active and Fossil Suture Zones, WIHG, Abstract, p. 131.

Piper, J.D.A., 1975. The Palaeomagnetism of Precambrian igneous and sedimentary rocks of the Orange River belt in South Africa and South West Africa. Geophysical Journal. Royal Astronomical Society 40 (3), 313–344.

Piper, J.D.A., 1982. The Precambrian palaeomagnetic record: the case for the Proterozoic supercontinent. Earth and Planetary Science Letters 59, 61–89.

Piper, J.D.A., 2010. Protopangaea: Palaeomagnetic definition of Earth's oldest (mid-Archaean-Palaeoproterozoic) supercontinent. Journal of Geodynamics 50, 154–165.

Piper, J.D.A., Zhang, J., Baochung, H., Roberts, A.P., 2011. Palaeomagnetism of Precambrian dyke swarms in the North China Shield: The 1.8 Ga LIP event and crustal

consolidation in late Palaeoproterozoic times. Journal of Asian Earth Sciences 41, 504–524.
Pradhan, V.R., Meert, J.G., Pandit, M.K., Kamenov, G., Mondal, E.A., 2012. Tectonic evolution of the Precambrian Bundelkhand craton, central India: insights from paleomagnetic and geochronologic studies on the mafic dyke swarms. Precambrian Research 198–199, 51–76.
Radhakrishna, T., Joseph, M., 1996a. Proterozoic palaeomagnetism of the mafic dyke swarms in the high-grade region of south India. Precambrian Research 76, 31–46.
Radhakrishna, T., Joseph, M., 1996b. Late Precambrian (~850–800 Ma) palaeomagnetic pole of south Indian shield from the Harohalli alkaline dykes: geotectonic implications for the Gondwana reconstructions. Precambrian Research 80, 77–87.
Radhakrishna, T., Joseph, M., Krishnendu, N.R., Balasubramonian, G., 2003. Palaeomagnetism of mafic dykes in the Karnataka Craton: possible geodynamic implications. Memoir Geological Society of India 50, 193–224.
Radhakrishna, T., Krishnendu, N.R., Balasubramonian, G., 2007. Mafic dyke magmatism around the Cuddapah basin: age constraints, petrological characteristics and geochemical inference for a possible magma chamber on the southwestern margin of the basin. Journal of the Geological Society of India 70, 194–206.
Radhakrishna, T., Chandra, R., Srivatava, A.K., Balasubramonian, G., Balakrishna, S., 2009. Palaeoproterozoic mafic (dyke) activity in Bundelkhand craton, India and tectonic implications. In: Abstract (late) in the Proceedings of the 2nd Precambrian Crustal Growth and Tectonism, Bundelkhand University, Jhansi, India, LA4, 2009.
Radhakrishna, T., Krishnendu, N.R., Balasubramonian, G. Palaeoproterozoic Indian shield in the global continental assembly: evidences from palaeomagnetism of mafic dykes. Earth-Science Reviews, submitted for publication.
Ramakrishnan, M., 1990. Crustal development in southern Bastar central Indian craton. Geological Survey of India Special Publication 28, 44–66.
Ramakrishnan, M., Vaidyanadhan, 2008. Geology of India, vol. 1. Geological Society of India, Bangalore, 556pp.
Ramchandra, H.M., Mishra, V.P., Deshmukh, S.S., 1995. Mafic dykes in the Bastar Precambrian: study of the Bhanupratappur–Keskal mafic dyke swarm. In: Devaraju, T.C. (Ed.), Mafic Dyke Swarms of Peninsular India, vol. 33. Memoirs Geological Society of India, pp. 183–207.
Rao, J.M., 2004. The wide-spread 2 Ga dyke activity in the Indian shield—evidences from Bundelkhand mafic dyke swarm, Central India and their tectonic implications. Gondwana Research 7, 1219–1228.
Rao, V.V., Reddy, P.R., 2002. A Mesoproterozoic supercontinent: evidence from the Indian Shield. Gondwana Research 5, 53–62.
Rao, J.M., Rao, G.V.S.P., Widdowson, M., Kelley, S.P., 2005. Evolution of Proterozoic mafic dyke swarms of the Bundelkhand Granite Massif, Central India. Current Science 88, 502–506.
Rogers, J.J.W., 1996. A history of the continents in the past three billion years. Journal of Geology 104, 91–107.
Rogers, J.J.W., Santosh, M., 2002. Configuration of Columbia, a Mesoproterozoic supercontinent. Gondwana Research 5, 5–22.
Salminen, J., Pesonen, L.J., Reimold, W.U., Donadini, F., Gibson, R.L., 2009. Paleomagnetic and rock magnetic study of the Vredefort impact structure and the Johannesburg Dome, Kaapvaal Craton, South Africa – implications for the apparent polar wander path of the Kaapvaal Craton during the Mesoproterozoic. Precambrian Research 168, 167–184.
Sarkar, G., Corfu, F., Paul, D.K., McNaughton, N.J., Gupta, S.N., Bishui, P.K., 1993. Early Archean crust in Bastar craton, Central India – a geochemical and isotopic study. Precambrian Research 62, 127–137.
Sharma, K.K., Rahman, A., 2000. The early Archaean-Paleoproterozoic crustal growth of the Bundelkhand craton, northern Indian shield. In: Deb, M. (Ed.), Crustal Evolution and Metallogeny in the Northwestern Indian Shield. Narosa Publishing House, New Delhi, pp. 51–72.
Söderlund, U., Hofmann, A., Klausen, M.B., Olsson, J.R., Ernst, R.C., Persson, P., 2010. Towards a complete magmatic barcode for the Zimbabwe craton: Baddeleyite U–Pb dating of regional dolerite dyke swarms and sill complexes. Precambrian Research 183, 388–398.
Srivastava, R.K., Singh, R.K., 2003. Geochemistry of high-Mg mafic dykes from the Bastar craton: evidence of an Archean boninite like rock in intracratonic setting. Current Science 85, 808–812.
Srivastava, R.K., Singh, R.K., 2004. Trace element geochemistry and genesis of Precambrian Subalkaline mafic dikes from the central Indian craton: evidence for mantle metasomatism. Journal of Asian Earth Sciences 23, 373–389.
Srivastava, R.K., Hall, R.P., Verma, R., Singh, R.K., 1996. Contrasting mafic dykes of the Bastar craton, central India: petrological and geochemical characteristics. Journal of the Geological Society of India 48, 537–546.
Srivastava, R.K., Heaman, L.M., French, J.E., Ferreira Fil, C.E., 2011. Evidence for a Paleoproterozoic event of metamorphism in the Bastar craton, Central India: *P–T–t* constraints from mineral chemistry and U–Pb geochronology of mafic dykes. Episodes 3, 13–24.
Stein, H.J., Hannah, J.L., Zimmerman, A., Markey, R.J., Sarkar, S.C., Pal, A.B., 2004. A 2.5 Ga porphyry Cu–Mo–Au deposit at Malanjhkhand, central India: implications for late Archean continental assembly. Precambrian Research 134, 189–226.
Venkatesh, A.S., Pooruachandrarao, G.V.S., Prasadarao, N.T.V., Bhalla, M.S., 1987. Palaeomagnetic and geochemical studies on dolerite dykes from Tamil Nadu, India. Precambrian Research 34, 291–310.
Williams, H., Hoffman, P.F., Lewry, J.F., Monger, J.W.H., Rivers, T., 1991. Anatomy of north America: thematic geology portrayals of the continent. Tectonophysics 187, 117–134.
Wingate, M.T.D., Pisarevsky, S.A., Gladkochub, D.P., Donskaya, T.V., Konstantinov, K.M., Mazukabzov, A.M., Stanevich, A.M., 2009. Geochronology and palaeomagnetism of mafic igneous rocks in the Olenek Uplift, northern Siberia: implications for Mesoproterozoic supercontinents and paleogeography. Precambrian Research 170, 256–266, http://dx.doi.org/10.1016/j.precamres.2009.01.004.
Zhang, S., Li, Z.-X., Evans, D.A.D., Wu, H., Li, H., Dong, J., 2012. Pre-Rodinia supercontinent Nuna shaping up: a global synthesis with new paleomagnetic results from North China. Earth and Planetary Science Letters 353–354, 145–155.
Zhao, G., Cawood, P.A., Wilde, S.A., Sun, M., 2002. Review of global 2.1–1.8 orogens: implications for a pre-Rodinia supercontinent. Earth-Science Reviews 59, 125–162.
Zhao, G.C., Sun, M., Wilde, S.A., 2003. Correlations between the Eastern Block of the North China Craton and the South Indian Block of the Indian Shield: an Archean to Paleoproterozoic link. Precambrian Research 122, 201–233.
Zhao, G., Sun, W., Wilde, S.A., Li, S.Z., 2004. A Paleo-Mesoproterozoic supercontinent: assembly, growth and breakup. Earth-Science Reviews 67, 91–123.